Year II — Algebra

Based on lectures by Various Lecturers

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Chapter 1

Rings, subrings and fields

1.1 Definition of a Ring

Before being able to define a ring, we must first redefine *binary operations*:

Definition 1.1.1 (Binary Operation)**.** A *binary opeation* is a function that takes two elements as an argument and produces a single new output value. Examples include addition, multiplication and subtraction.

Remark 1.1.2. All rings form Abelian groups (review first year if unsure what this means)

Definition 1.1.3 (Ring)**.** A ring R is a set together with two binary operations (usually, but not always, written,$+\cdot$) called "addition" and "multiplication", such that $+$ makes R into an abelian group, and such that the following conditions hold:

i) **Existence of an Identity** - There exists an element $1 \in R$ such that $1 \cdot x = x \cdot 1 = x$, for all $x \in R$.

ii) **Associativity** - For any $x, y, z \in R$ we have $x(yz) = (xy)z$.

iii) **Distributivity** - For any $x, y, z \in R$ we have $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

1.2 Examples of Rings

– Let V be a finite dimensional vector space over $\mathbb{C}$. Let

$$\mathrm{End}(V) = \{f : V \to V \mid f \text{ is a linear map } \}.$$

An element of $\mathrm{End}(V)$ is usually called a (linear) endomorphism. We can add two such functions like we have seen before:

$$(f + g)(v) = f(v) + g(v).$$

Now we can't directly "multiply" two endomorphisms, because V does not have any multiplication (V is not in general a ring). However, we can compose two endomorphisms

$$(f \circ g)(v) = f(g(v)),$$

and composition is a binary operation on V. From Linear Algebra I, we already know that $\mathrm{End}(V)$ is an abelian group under addition. To check that this is a ring, we need to check the three properties:

Identity element is provided by the identity map $Id : V \to V$. Associativity and distributivity are also not hard to check. For example we need to establish that if $f, g, h \in \text{End}(V)$ we have

$$f \circ (g \circ h) = (f \circ g) \circ h.$$

That is, for every vector $v \in V$ we have that $(f \circ (g \circ h))(v) = ((f \circ g) \circ h)(v)$. For the definition of the "multiplication" we have that the left hand side is equal to

$$(f \circ (g \circ h))(v) = f((g \circ h)(v)) = f(g(h(v))$$

and

$$((f \circ g) \circ h)(v) = (f \circ g)(h(v)) = f(g(h(v))$$

That is the two endomorphisms are having the same effect on any vector v of V. That is they are the sane maps. Similarly one checks distributivity.

- $\mathbb{N}$ is not a ring ($1 - 2 \notin \mathbb{N}$)
- $\mathbb{Z}$ is a ring but not a field: $\frac{1}{2} \notin \mathbb{Z}$.
- $\mathbb{Q}[x]$ is a ring but not a field: $\frac{1}{x} \notin \mathbb{Q}[x]$.
- $\mathrm{M}_n(\mathbb{Q})$ is a ring but not a field: there is no way we can make sense of division. For example, $\frac{1}{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}$ is meaningless since that matrix does not have an inverse.
- The set $C(\mathbb{R})$ is a ring but not a field: the constant function $1 : \mathbb{R} \to \mathbb{R}$ belongs to the set, and so does the identity function Id defined by $\text{Id}(x) = x$ for all $x \in \mathbb{R}$. However, the function $\frac{1}{\sqrt{d}}$ is not continuous at $x = 0$, so does not belong to $C(\mathbb{R})$.

Method 1.2.1 (Determining a Ring)**.** To determine a ring, one should first check that the ring forms an abelian group with respect to the 'addition' operation and that the following requirements hold for the 'multiplication' operation:

i) **Existence of an Identity** - There exists an element $1 \in R$ such that $1 \cdot x = x \cdot 1 = x$, for all $x \in R$.

ii) **Associativity** - For any $x, y, z \in R$ we have $x(yz) = (xy)z$.

iii) **Distributivity** - For any $x, y, z \in R$ we have $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

1.3 The integers modulo n

Definition 1.3.1 (Equivalence Relation)**.** A (binary) relation $\sim$, on a set S is said to be an equivalence relation, if and only if it is reflexive, symmetric and transitive. That is, for all a, b, and c in the set S we have:

i) $a \sim a$ (reflexivity).

ii) $a \sim b$ if and only if $b \sim a$ (symmetry).

iii) If $a \sim b$ and $b \sim c$ then $a \sim c$ (transitivity).

Definition 1.3.2 (Equivalence Class)**.** The equivalence class of a under the given relation $\sim$, denoted by $[a]$ (or $\bar{a}$) is defined as

$$[a] = \{x \in S : x \sim a\}.$$

Note that for two $a, b \in S$ with $a \sim b$ we have $[a] = [b]$. In particular when we write $[a]$ we call a the representative of the class $[a]$. Clearly an equivalence class may have more than one representative.

From these structures we can construct another set:

$$S/\sim := \{[a] : a \in S\}$$

That is, this is the set of equivalence classes on the set S with respect to the given equivalence relation $\sim$.

Definition 1.3.3 (Integers Modulon n)**.** The integers modulo n. Let $n \in \mathbb{N}$ be a natural number. Let us consider the set of integer numbers $\mathbb{Z}$ and define the following relation: two integers a and b are equivalent (i.e. $a \sim b$ in the above notation) if n divides $a - b$. The most common notation for this is $a \equiv b \pmod{n}$.

Proof of 'the relation is an equivalence relation'.

- it is reflexive as $a - a = 0$ and hence divisible by n,
- it is symmetric as n divides $a - b$ if and only if n divides $b - a = -(a - b)$.
- And finally for the last condition we consider $a, b, c \in \mathbb{Z}$ and further assume that n divides $a - b$ and $b - c$, that is $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$. That is, $a - b = kn$ and $b - c = rn$ for some $k, r \in \mathbb{Z}$. But then $a - c = a - b + b - c = (k + r)n$ and hence n divides $a - c$ i.e. $a \equiv c \pmod{n}$.

□

Definition 1.3.4 (Residue Class)**.** For an integer a we denote by $\bar{a}$ (this is the most commonly used notation) the equivalence class of a, that is

$$\bar{a} = \{x \in \mathbb{Z} : n \text{ divides } x - a\} = \{x \in \mathbb{Z} : x = a \pmod{n}\}$$

Now note that when we divide an integer by n we will always get a remainder among the numbers

$$0, 1, , \ldots, n - 1.$$

That is, every integer belongs to one of the following equivalence classes:

$$\{\overline{0}, \overline{1}, \ldots, \overline{n-1}\},$$

We will denote this set (of equivalence classes) by $\mathbb{Z}/n$. The elements of $\mathbb{Z}/n$ are called residue classes.

Remark 1.3.5. Make sure that if you ever write $\mathbb{Z}_n$ that you explain that you mean $\mathbb{Z}/n$ as it can cause confusion.

1.4 Subrings and Fields

Definition 1.4.1 (Subring)**.** A *subring* S of a ring R is a set $S \subseteq R$ such that

i) $0 \in S, 1 \in S$ where 0 is the 'additive' identity and 1 it the 'multiplicative identity',

ii) $a, b \in S \Rightarrow a + b \in S$,

iii) $a, b \in S \Rightarrow ab \in S$,

iv) $a \in S \Rightarrow -a \in S$.

Remark 1.4.2 (Intuitive definition of Subring)**.** We can shorten this into an informal definition of 'A subring contains 0_R, 1_R and is closed under addition, subtraction and multiplication.

Remark 1.4.3 (Subrings are Rings)**.** A subring of any ring is a ring itself and is equipped with addition and multiplication of the original ring.

Method 1.4.4 (Showing a Subring)**.** We can quickly show a subring by showing:

- the additive and multiplicative identities exist in the set,
- it's closed under addition
- it's closed under multiplication
- it contains additive inverses.

Method 1.4.5 (Determining a Ring quickly)**.** Rather than displaying a set satisfies all of the properties of a ring, it is sufficient to show that the set is a subring of a ring. It is therefore a ring itself.

Definition 1.4.6 (Field)**.** A ring R is called a field if:

i) R is commutative,

ii) $1 \neq 0$ in R (i.e., R has at least two elements),

iii) For any $a \in R$ such that $a \neq 0$, there exists a $b \in R$ such that $ab = 1$ (b is called the inverse of a).

Chapter 2

Arithmetic in $\mathbb{Z}$ and $F[x]$

2.1 Divisibility in $\mathbb{Z}$

Definition 2.1.1 (Division)**.** In a commutative ring R, an element $a \in R$ is said to divide an element $b \in R$, written $a \mid b$, if there exists an $x \in R$ such that

$$b = ax.$$

We will sometimes write $a \nmid b$ if a does not divide b.

Remark 2.1.2. It is easy to show from the definition that if $a \mid b$ and $a \mid c$ for $a, b, c \in R$, then $a \mid (bx+cy)$, for any $x, y \in R$. In particular, if a divides two elements, then it divides the sum and difference of those two elements.

Lemma 2.1.3 (Well-Ordering Principle (WOP))**.** Let S be a non-empty subset of $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$. Then S has a smallest element.

Proposition 2.1.4. Let $a \in \mathbb{Z}, b \neq 0$. Then there exist in tegers q and r such that

$$a = q \cdot b + r, \quad \text{and} \quad 0 \leq r < |b|.$$

(q is called the quotient and r the remainder).

Proof. Insert proof here □

Example 2.1.5. Find the integer quotient and remainder of 435 divided by 43 in $\mathbb{Z}$. Long division with remainder gives:

$$435 = 43 \cdot 10 + 5,$$

and $5 < 43$, so we stop. So the integer quotient is $q = 10$ and the remainder is $r = 5$.

Theorem 2.1.6 (Euclidean Algorithm in $\mathbb{Z}$)**.** Let $a, b \in \mathbb{Z}$, not both zero. Then $\gcd(a, b)$ exists and can be computed by the Euclidean algorithm (given in the proof). Moreover, if $d = \gcd(a, b)$, then there exist $x, y \in \mathbb{Z}$ such that

$$ax + by = d.$$

Proof. Insert proof here □

Example 2.1.7. Example 2.5. We use the Euclidean Algorithm to compute $d = \gcd(455, 1235)$ and find integers x, y such that $d = 455x + 1235y$.

Solution: Division with remainder:

$$\begin{aligned} 1235 &= 2 \cdot 455 + 325 \\ 455 &= 1 \cdot 325 + 130 \\ 325 &= 2 \cdot 130 + 65 \\ 130 &= 2 \cdot 65 + 0. \end{aligned}$$

When the remainder is 0 , we stop. The gcd will be the last non-zero remainder, here $d = 65$. Solving for the remainders, substituting back and simplifying, we get

$$\begin{aligned} 65 &= 325 - 2 \cdot 130 \\ &= 325 - 2(455 - 1 \cdot 325) \\ &= \cdots \\ &= 455 \cdot (-8) + 1235 \cdot 3. \end{aligned}$$

so $x = -8$ and $y = 3$.

Lemma 2.1.8 (Primes that divide products divide at least one factor)**.** If p is a prime and $p \mid ab$ for some $a, b \in \mathbb{Z}$, then $p \mid a$ or $p \mid b$. (In other words, p prime $\Rightarrow (\forall a, b \in \mathbb{Z}, p|ab \Rightarrow p|a$ or $p \mid b)$.)

Theorem 2.1.9 (Fundamental Theorem of Arithmetic)**.** Let $n \in \mathbb{N}, n > 1$. Then n can be written as a product of primes in a unique way (up to the order of the factors). More precisely, there are distinct primes $p_1, \ldots, p_r$ and $e_1, \ldots, e_r \in \mathbb{N}$ such that

$$n = p_1^{e_1} \cdots p_r^{e_r},$$

and if $n = q_1^{f_1} \cdots q_s^{f_s}$ for some distinct primes $q_1, \ldots, q_s$ and $f_1, \ldots, f_s \in \mathbb{N}$, then $r = s$, and up to re-ordering, $p_i = q_i$ and $e_i = f_i$ for all i.

2.2 Divisibility in $F[x]$

Definition 2.2.1 (Degree)**.** For $f(x) = a_0 + a_1 x + \cdots + a_n x^n \in F[x]$, its *degree* is

$$\deg f := \begin{cases} \max\{i \mid a_i \neq 0\} & \text{if } f(x) \neq 0, \\ -\infty & \text{if } f(x) = 0. \end{cases}$$

Then we have the following basic properties for all $f(x), g(x) \in F[x]$:

- $\deg(fg) = \deg f + \deg g$,
- $\deg(f + g) \leq \max\{\deg f, \deg g\}$. Note that if $f(x) \neq g(x)$, then equality holds

Remark 2.2.2. We define $\deg(f) = -\infty$ if $f(x) = 0$ to preserve the properties of degree in the definition.

If $f = 0$ and $\deg 0 = \deg(0 \cdot g) = \deg 0 + \deg g$, then we can't allow $\deg 0$ to be in $\mathbb{Z}$, but it would work if $\deg 0 = \pm\infty$. However, if also $\deg g = \deg(0 + g) \leq \max\{\deg 0, \deg g\}$, for all g, then we can't allow $\deg 0 = \infty$, but it does work if we set $\deg 0 = -\infty$.

Proposition 2.2.3. Let $f(x), g(x) \in F[x]$ with $g(x) \neq 0$. Then there exist polynomials $q(x), r(x) \in F[x]$ with $\deg r < \deg g$ such that

$$f(x) = q(x)g(x) + r(x).$$

Example 2.2.4 (Long division of $f(x) = x^3 + x^2 - 3x - 3$ by $g(x) = x^2 + 3x + 2$, with remainder)**.**

term to get rid of		$x - 2$
x^3	$x^2 + 3x + 2$	$x^3 + x^2 - 3x - 3$ $-\left(x^3 + 3x^2 + 2x\right)$
$-2x^2$		$-2x^2 - \left(-2x^2 - 6x - 3 \ -7\right)$
		$x + 1$

We have obtained a remainder of degree 1 , which is smaller than $\deg g$, so we stop:

$$f(x) = (x - 2)g(x) + x + 1.$$

Thus $q(x) = x - 2$ and $r(x) = x + 1$. Think of $q(x)$ as the "quotient" and $r(x)$ as the "remainder".

Proof. insert proof here □

2.3 GCD's and the Euclidean Algorithm in $F[x]$